Lecture 10

The Gelfand–Tsetlin basis of S_n

Objectives. *Block III opens with the representations of S_n. Restriction along $S_1 \subset \cdots \subset S_n$ is multiplicity-free, so every irreducible splits canonically into lines. Choosing a non-zero vector on each line gives a* Gelfand–Tsetlin basis*; the individual vectors depend on these scalar choices. The Jucys–Murphy elements $X_1, \ldots, X_n$ commute and generate the algebra of elements acting diagonally in the GZ bases of all irreducibles simultaneously. Their joint eigenvalues distinguish the lines. Lectures 11 and 12 find which occur.*

By the end of this lecture you will be able to:

- Prove that, for $n \geq 2$, every permutation in S_n is conjugate to its inverse by a permutation fixing n, and deduce that each irreducible representation of S_n restricts to S_{n-1} without multiplicity.
- Construct the Gelfand–Tsetlin basis of an irreducible representation of S_n, and prove that $\mathrm{GZ}(n)$ is a maximal commutative subalgebra of $\mathbb{C}[S_n]$, diagonal in that basis.
- Prove that $X_1, \ldots, X_n$ generate $\mathrm{GZ}(n)$, deduce that weights determine Gelfand–Tsetlin vectors up to scalars, and compute weights from character values.
- Derive the relations $s_i X_i + 1 = X_{i+1} s_i$ and $s_i X_k = X_k s_i$ for $1 \leq i \leq n-1$ and $1 \leq k \leq n$ with $k \neq i, i+1$.

10.1 Multiplicity-free branching

The subgroup $S_k \leq S_n$ is the stabiliser of $k+1, \ldots, n$, so

$$S_1 \subset S_2 \subset \cdots \subset S_n,$$

and $\mathbb{C}[S_k] \subseteq \mathbb{C}[S_n]$. By the Gelfand criterion (Theorem 7.20), each irreducible representation of S_n restricts to S_{n-1} without multiplicity if and only if $\mathbb{C}[S_n]^{S_{n-1}}$ is commutative. A basis of this algebra is the sums over the S_{n-1}-*classes* $\{h\sigma h^{-1} : h \in S_{n-1}\}$ of S_n (§ 7.5).

Lemma 10.1 (The S_{n-1}-classes)**.** *Let $n \geq 2$. Two permutations $\sigma, \sigma' \in S_n$ are conjugate by an element of S_{n-1} if and only if they have the same cycle type and their cycles through n have the same length.*

Proof. Only if. If $\tau \in S_{n-1}$, then $\tau\sigma\tau^{-1}$ has the cycles of σ with each entry a replaced by $\tau(a)$ (Theorem 2.16(i)). As $\tau(n) = n$, the cycle through n goes to the cycle through n.

If. Write the cycles of σ and σ', fixed points included, the cycle through n first and beginning with n, and the other cycles with equal lengths aligned:

$$\sigma = (n\, a_2 \ \ldots\ a_k)(b_1 \ \ldots\ b_l)\cdots,$$
$$\sigma' = (n\, a'_2 \ \ldots\ a'_k)(b'_1 \ \ldots\ b'_l)\cdots.$$

The map τ sending each entry to the one below it is a permutation with $\tau(n) = n$, and $\tau\sigma\tau^{-1} = \sigma'$ by Theorem 2.16(i). ∎

The inverse of a cycle is the reversed cycle, so σ^{-1} has the cycle type of σ, and the same length of cycle through n. By the lemma, σ and σ^{-1} are conjugate by an element fixing n, as Lecture 2 promised.

Example 10.2 (An element fixing 4)**.** In S_4, aligning

$$(1\,2\,4\,3) = (4\,3\,1\,2),$$
$$(1\,2\,4\,3)^{-1} = (4\,2\,1\,3)$$

gives $\tau = (2\,3)$, which swaps 2 and 3 and fixes 1 and 4.

The *linear inversion map* sends $a = \sum_\sigma a_\sigma \sigma \in \mathbb{C}[S_n]$ to $\iota(a) = \sum_\sigma a_\sigma \sigma^{-1}$.

Theorem 10.3 (Multiplicity-free branching)**.** *Let $n \geq 2$.*

(i) $\iota(a) = a$ *for every* $a \in \mathbb{C}[S_n]^{S_{n-1}}$.

(ii) $\mathbb{C}[S_n]^{S_{n-1}}$ *is commutative.*

(iii) For every $U \in \operatorname{Irr}(S_n)$, $\operatorname{Res}^{S_n}_{S_{n-1}} U$ *is multiplicity-free.*

Proof. (i) The coefficients of a are constant on S_{n-1}-classes, and σ^{-1} lies in the class of σ (Lemma 10.1). So $a_{\sigma^{-1}} = a_\sigma$, and

$$\iota(a) = \sum_\sigma a_\sigma\, \sigma^{-1} = \sum_\sigma a_{\sigma^{-1}}\, \sigma = a.$$

(ii) The map ι is linear, and $(\sigma\rho)^{-1} = \rho^{-1}\sigma^{-1}$, so $\iota(ab) = \iota(b)\iota(a)$ on basis elements, hence for all a and b. If a and b lie in $\mathbb{C}[S_n]^{S_{n-1}}$, so does ab, and by (i)

$$ab = \iota(ab) = \iota(b)\iota(a) = ba.$$

(iii) The centraliser algebra is commutative, so Theorem 7.20 gives multiplicity-free restriction. ∎

10.2 Gelfand–Tsetlin bases and $\mathrm{GZ}(n)$

Multiplicity-free restriction means each irreducible occurs at most once, so each occurring type contributes one irreducible isotypic component. These components are intrinsic; successive restrictions will give a subspace for each path in the branching graph.

Definition 10.4 (Branching graph, paths)**.** The *branching graph* of $S_1 \subset \cdots \subset S_n$ has as vertices the irreducibles $U \in \operatorname{Irr}(S_k)$, $1 \leq k \leq n$, and an edge $U \to U'$ from $U \in \operatorname{Irr}(S_{k-1})$ to $U' \in \operatorname{Irr}(S_k)$ when U occurs in $\operatorname{Res}^{S_k}_{S_{k-1}} U'$, exactly once by Theorem 10.3. A *path* to $V \in \operatorname{Irr}(S_n)$ is a chain

$$\gamma = (U_1 \to U_2 \to \cdots \to U_n),$$
$$U_k \in \operatorname{Irr}(S_k), \quad U_n = V.$$

The paths to all $V \in \operatorname{Irr}(S_n)$ are the *paths of length* n in this course: each has n vertices and $n - 1$ arrows.

The group S_1 is trivial, so $U_1 = \mathbf{1}$. By Example 7.21, E has the two paths

$$\mathbf{1} \to \mathbf{1} \to E, \qquad \mathbf{1} \to \operatorname{sgn} \to E.$$

The path subspaces. Let $V \in \operatorname{Irr}(S_n)$ and γ a path to V. Put $W_n = V$, and for $k = n, \dots, 2$ let W_{k-1} be the U_{k-1}-isotypic component of $\operatorname{Res}^{S_k}_{S_{k-1}} W_k$ (Proposition 7.14). Let $V_\gamma = W_1$. Each W_k depends only on the tail $(U_k, \dots, U_n)$.

Definition 10.5 (Gelfand–Tsetlin algebra)**.** The *Gelfand–Tsetlin algebra* is

$$\operatorname{GZ}(n) = \left\langle Z(\mathbb{C}[S_1]), \dots, Z(\mathbb{C}[S_n]) \right\rangle \subseteq \mathbb{C}[S_n].$$

The brackets mean generation as a unital algebra, not an inner product or subgroup generation. We abbreviate Gelfand–Tsetlin to GZ.

We use the matrix-block map of Theorem 7.9, $\Phi\colon \mathbb{C}[S_n] \to \bigoplus_{V \in \operatorname{Irr}(S_n)} \operatorname{End}(V)$, $\Phi(a) = (\rho_V(a))_V$.

Theorem 10.6 (Gelfand–Tsetlin bases)**.** *Let $V \in \operatorname{Irr}(S_n)$.*

(i) Each W_k is isomorphic to U_k, and V_γ is a line. V is the direct sum of the V_γ over the paths γ to V. Any choice of non-zero vectors $v_\gamma \in V_\gamma$ is a basis of V, and $\dim V$ is the number of paths to V.

(ii) $\operatorname{GZ}(n)$ is commutative, and each $z \in Z(\mathbb{C}[S_k])$, $k \le n$, acts on V_γ by the scalar $\omega_{U_k}(z)$ (Definition 9.4).

(iii) Φ maps $\operatorname{GZ}(n)$ onto the elements of $\bigoplus_V \operatorname{End}(V)$ that are diagonal in such bases, independently of the choices. So $\operatorname{GZ}(n)$ is a maximal commutative subalgebra of $\mathbb{C}[S_n]$, of dimension the number of paths of length n.

Proof. (i) *Irreducible.* By downward induction on k: $W_n = V$, which is U_n; and if $W_k \cong U_k$, then $\operatorname{Res}^{S_k}_{S_{k-1}} W_k$ contains U_{k-1} exactly once (Theorem 10.3), so $W_{k-1} \cong U_{k-1}$. As S_1 is trivial, $V_\gamma = W_1$ is a line.

The sum. $\operatorname{Res}^{S_k}_{S_{k-1}} W_k$ is the direct sum of its isotypic components, one for each edge into U_k (Proposition 7.14). Splitting V, then each W_{n-1}, and so on, writes V as the direct sum of the V_γ, each determined by V and γ.

(ii) For $j \le k$, $Z(\mathbb{C}[S_k])$ commutes with $\mathbb{C}[S_j] \subseteq \mathbb{C}[S_k]$, so the generators of $\operatorname{GZ}(n)$ commute pairwise. A central z of $\mathbb{C}[S_k]$ acts on $W_k \cong U_k$ by $\omega_{U_k}(z)$ (Theorem 7.11(ii), Proposition 7.5(ii)), and $V_\gamma \subseteq W_k$.

(iii) *Diagonal.* By (ii), every element of $\operatorname{GZ}(n)$ acts diagonally in each basis $\{v_\gamma\}$.

Onto. For a path $\gamma = (U_1 \to \cdots \to U_n)$, let

$$P_\gamma = e_{U_1} e_{U_2} \cdots e_{U_n} \in \operatorname{GZ}(n),$$

with $e_{U_k} \in Z(\mathbb{C}[S_k])$ the central idempotent of U_k. By (ii) and Proposition 7.12, e_{U_k} acts on the line $V'_{\gamma'}$ of a path $\gamma' = (U'_1 \to \cdots \to U'_n)$ by 1 if $U'_k = U_k$ and by 0 otherwise. So P_γ acts by 1 on V_γ and by 0 on every other line: $\Phi(P_\gamma)$ is the diagonal matrix unit at γ, and these span the diagonal elements.

Maximal. If $a \in \mathbb{C}[S_n]$ commutes with $\operatorname{GZ}(n)$, then $\Phi(a)$ commutes with every $\Phi(P_\gamma)$, so it preserves every line V_γ and is diagonal. So $\Phi(a) = \Phi(b)$ for some $b \in \operatorname{GZ}(n)$, and $a = b$, as Φ is injective (Theorem 7.9). ∎

Definition 10.7 (Gelfand–Tsetlin basis)**.** A *Gelfand–Tsetlin basis* of V chooses one non-zero vector v_γ on each line V_γ. A non-zero vector on such a line is a *GZ vector*; each basis vector can be rescaled independently by a non-zero scalar.

10.3 Jucys–Murphy elements

We now use commuting elements of $\mathrm{GZ}(n)$ to give numerical labels to the path lines.

Definition 10.8 (Jucys–Murphy elements)**.** In $\mathbb{C}[S_n]$, let $X_1 = 0$ and, for $2 \le k \le n$,

$$X_k = (1\,k) + (2\,k) + \cdots + (k{-}1\;\,k).$$

Let $\tau_k = \sum_{1 \le i < j \le k} (i\,j)$, the class sum of the transpositions of S_k, with $\tau_0 = 0$ and $\tau_1 = 0$.

The transpositions of S_k not in S_{k-1} are those that move k, so

$$X_k = \tau_k - \tau_{k-1}.$$

As $\tau_k \in Z(\mathbb{C}[S_k])$ (Theorem 7.11(i)), $X_k \in \mathrm{GZ}(n)$, and the X_k commute (Theorem 10.6(ii)).

Definition 10.9 (Weights)**.** For a GZ vector v, write $X_k v = a_k v$. Its *weight* is $\alpha(v) = (a_1, \ldots, a_n)$, and $\mathrm{Spec}(n)$ is the set of weights of the GZ vectors of all $V \in \mathrm{Irr}(S_n)$.

Proposition 10.10 (Weights from characters)**.** *Let $v \in V_\gamma$ be a GZ vector, $\gamma = (U_1 \to \cdots \to U_n)$. Then $a_1 = 0$ and, for $2 \le k \le n$,*

$$a_k = \omega_{U_k}(\tau_k) - \omega_{U_{k-1}}(\tau_{k-1}),$$

where, for $2 \le k \le n$, $U \in \mathrm{Irr}(S_k)$ and a transposition t of S_k,

$$\omega_U(\tau_k) = \frac{k(k-1)}{2} \cdot \frac{\chi_U(t)}{\dim U},$$

and $\omega_{U_1}(\tau_1) = 0$. So a_k depends only on the edge $U_{k-1} \to U_k$.

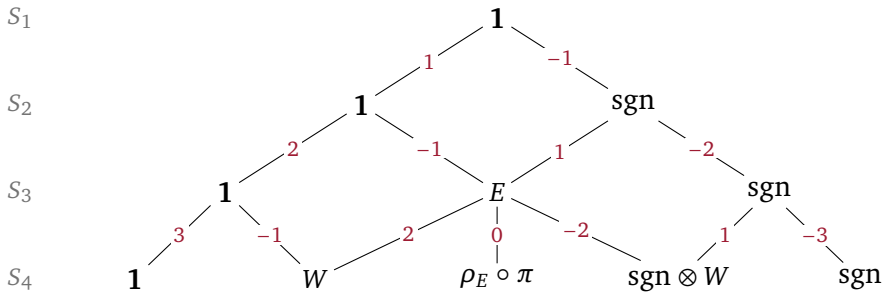
Proof. By Theorem 10.6(ii), τ_k acts on v by $\omega_{U_k}(\tau_k)$, and $X_k = \tau_k - \tau_{k-1}$. S_k has $k(k-1)/2$ transpositions, so Lemma 9.5(i) gives $\omega_U(\tau_k)$. ∎

Example 10.11 (The branching graph of S_4)**.** Recall from § 6.6 the coordinate-sum-zero representation W, its sign twist, and the lift $\rho_E \circ \pi$; here $\pi : S_4 \to S_3$ is the action on three pairings (§ 2.6), and ρ_E is the action on the plane E (§ 5.5). On e, $(1\,2)$ and $(1\,2\,3)$, the restricted character χ_W has values $(3, 1, 0)$, the sum of those of $\mathbf{1}$ and E (§ 6.6 and Example 6.4). Character determination gives the first restriction below. In the same way, the three irreducibles of degree greater than 1 restrict to S_3 as

$$\begin{aligned} \mathrm{Res}^{S_4}_{S_3} W &\cong \mathbf{1} \oplus E, \\ \mathrm{Res}^{S_4}_{S_3} (\mathrm{sgn} \otimes W) &\cong \mathrm{sgn} \oplus E, \\ \mathrm{Res}^{S_4}_{S_3} (\rho_E \circ \pi) &\cong E. \end{aligned}$$

The linear representations $\mathbf{1}$ and sgn restrict to $\mathbf{1}$ and sgn.

So, with Example 7.21, the branching graph of $S_1 \subset \cdots \subset S_4$ is



with 1, 3, 2, 3, 1 paths to $\mathbf{1}$, W, $\rho_E \circ \pi$, $\operatorname{sgn} \otimes W$, sgn, their degrees. By Proposition 10.10, from

S_2	$\mathbf{1}$	sgn			
$\omega_U(\tau_2)$	1	-1			
S_3	$\mathbf{1}$	E	sgn		
$\omega_U(\tau_3)$	3	0	-3		
S_4	$\mathbf{1}$	W	$\rho_E \circ \pi$	$\operatorname{sgn} \otimes W$	sgn
$\omega_U(\tau_4)$	6	2	0	-2	-6

each edge into level k carries its a_k, as drawn. For example, the edge $E \to W$ has label $\omega_W(\tau_4) - \omega_E(\tau_3) = 2 - 0 = 2$. The labels down a path, after $a_1 = 0$, give its weight: $(0, 1, -1, 2)$ for $\mathbf{1} \to \mathbf{1} \to E \to W$.

10.4 Generation

To prove that the Jucys–Murphy elements generate $\mathrm{GZ}(n)$, we first describe the centraliser algebra at one step of the chain. Generation will then show that their joint eigenvalues separate the path lines.

For $\sigma \in S_n$, let $\mathcal{K}(\sigma)$ be its S_{n-1}-class and $\widehat{\mathcal{K}}(\sigma)$ the sum of its elements; these sums are a basis of $\mathbb{C}[S_n]^{S_{n-1}}$. Conjugation keeps the number of points a permutation moves. If σ fixes n, then $\mathcal{K}(\sigma)$ is its class in S_{n-1}, and $\widehat{\mathcal{K}}(\sigma) \in Z(\mathbb{C}[S_{n-1}])$.

Proposition 10.12 (Olshanskii, for $S_{n-1} \le S_n$). *Let $n \ge 2$. Then*

$$\mathbb{C}[S_n]^{S_{n-1}} = \big\langle Z(\mathbb{C}[S_{n-1}]), X_n \big\rangle.$$

In particular, $Z(\mathbb{C}[S_n]) \subseteq \langle Z(\mathbb{C}[S_{n-1}]), X_n \rangle$.

Proof. Let $A = \langle Z(\mathbb{C}[S_{n-1}]), X_n \rangle$.

The inclusion in the centraliser algebra. Each $h \in S_{n-1}$ commutes with $Z(\mathbb{C}[S_{n-1}])$, and, as h fixes n,

$$h X_n h^{-1} = \sum_{i<n} (h(i)\ n) = X_n.$$

Thus the generators of A lie in the centraliser algebra, and so does A.

Products through n. Let $a_1, \dots, a_u < n$ be distinct. Right to left, the product

$$(a_1\ n)(a_2\ n) \cdots (a_u\ n)$$

sends

$$n \mapsto a_u, \qquad a_j \mapsto a_{j-1}\ (1 < j \le u), \qquad a_1 \mapsto n$$

so it is the $(u+1)$-cycle $(n\, a_u \ \dots\ a_2\, a_1)$; for example,

$$(1\,4)(2\,4) = (4\,2\,1).$$

Each $(u+1)$-cycle through n arises from exactly one such sequence. If the a_j are not distinct, the product moves only points among $n, a_1, \dots, a_u$, at most u of them. So

$$X_n^u = \big(\text{the sum of the } (u+1)\text{-cycles through } n\big) + \big(\text{terms moving at most } u \text{ points}\big).$$

The induction. We show $\widehat{\mathcal{K}}(\sigma) \in A$ by strong induction on the number d of points σ moves. If σ fixes n, $\widehat{\mathcal{K}}(\sigma) \in Z(\mathbb{C}[S_{n-1}]) \subseteq A$; this covers $d = 0$. Otherwise $\sigma = c\pi$, with c its cycle through n, of length $u+1 \ge 2$, and $\pi \in S_{n-1}$ the product of its other cycles, moving $d-u-1$ points. Let $C = \widehat{\mathcal{K}}(\pi) \in Z(\mathbb{C}[S_{n-1}])$, and expand $X_n^u C$ into terms xy, with x a product as above and y in the class of π.

Leading terms. The term xy moves at most

$$(u+1) + (d-u-1) = d$$

points, and exactly d only if x is a $(u+1)$-cycle through n moving no point that y moves. Then xy has the cycle type of σ and a cycle of length $u+1$ through n, so it lies in $\mathcal{K}(\sigma)$ (Lemma 10.1).

Coefficient one. Let $\sigma' \in \mathcal{K}(\sigma)$. It moves d points, so only leading terms can equal it. Its cycle through n determines the sequence for x, and the other cycles determine y. As σ' has the cycle type of σ, those other cycles have the cycle type of π, so y lies in the class of π in S_{n-1} (Theorem 2.16(ii)). Thus σ' occurs exactly once. So

$$X_n^u C = \widehat{\mathcal{K}}(\sigma) + r,$$

with r a combination of permutations moving fewer than d points.

The remainder. As $X_n^u C$ and $\widehat{\mathcal{K}}(\sigma)$ lie in $\mathbb{C}[S_n]^{S_{n-1}}$, so does r, and its coefficients are constant on S_{n-1}-classes: r is a combination of sums of classes moving fewer than d points. These lie in A by induction, and $X_n^u C \in A$; so $\widehat{\mathcal{K}}(\sigma) \in A$. Every S_{n-1}-class sum is therefore in A; these sums form a basis of the centraliser algebra. This proves the reverse inclusion, and hence $A = \mathbb{C}[S_n]^{S_{n-1}}$.

The centre lies in the centraliser. Every central element of $\mathbb{C}[S_n]$ commutes with S_{n-1}, so $Z(\mathbb{C}[S_n]) \subseteq \mathbb{C}[S_n]^{S_{n-1}}$. ∎

Example 10.13 (S_2 in S_3 again)**.** In S_3,

$$(1\,3)(2\,3) = (1\,3\,2), \qquad (2\,3)(1\,3) = (1\,2\,3)$$

so

$$\begin{aligned} X_3^2 &= (1\,3)^2 + (2\,3)^2 + (1\,3)(2\,3) + (2\,3)(1\,3) \\ &= 2e + (1\,2\,3) + (1\,3\,2). \end{aligned}$$

Example 7.21's basis of $\mathbb{C}[S_3]^{S_2}$ is therefore

$$e, \qquad (1\,2), \qquad X_3, \qquad X_3^2 - 2e,$$

with $(1\,2) \in Z(\mathbb{C}[S_2])$.

Theorem 10.14 (Jucys–Murphy generation). $\mathrm{GZ}(n) = \langle X_1, \dots, X_n \rangle$.

Proof. By induction on n. $\mathrm{GZ}(1)$ is $\mathbb{C}e$, which is $\langle X_1 \rangle$. Let $n \geq 2$. Each X_k lies in $\mathrm{GZ}(n)$ (§ 10.3). Conversely, $\mathrm{GZ}(n)$ is generated by $\mathrm{GZ}(n-1)$ and $Z(\mathbb{C}[S_n])$, and by Proposition 10.12 and induction,

$$\begin{aligned} Z(\mathbb{C}[S_n]) &\subseteq \langle Z(\mathbb{C}[S_{n-1}]), X_n \rangle \\ &\subseteq \langle \mathrm{GZ}(n-1), X_n \rangle \\ &= \langle X_1, \dots, X_n \rangle. \end{aligned}$$ ∎

Corollary 10.15 (Weights separate). *If GZ vectors $v \in V_\gamma$ and $v' \in V'_{\gamma'}$, of irreducibles V and V' of S_n, have the same weight, then $V = V'$ and $\gamma = \gamma'$. So $\gamma \mapsto \alpha(v_\gamma)$ is a bijection from the paths of length n onto* $\mathrm{Spec}(n)$.

Proof. By Theorem 10.14, $P_\gamma = f(X_1, \dots, X_n)$ for a polynomial f with complex coefficients, and the X_k commute. So P_γ acts on v and on v' by the same scalar $f(a_1, \dots, a_n)$. It is 1 on v, so it is 1 on v', and $\gamma' = \gamma$ by the proof of Theorem 10.6(iii). ∎

In Example 10.11, the ten paths of length 4 have ten different weights.

Example 10.16 (Guard: a diagonal element need not separate). An element of $\mathrm{GZ}(n)$ acts diagonally, but its eigenvalues need not separate GZ vectors. In S_3,

$$\tau_3 = X_1 + X_2 + X_3 \in \mathrm{GZ}(3),$$

and τ_3 acts on E by

$$\omega_E(\tau_3) = 3 \cdot \frac{0}{2} = 0,$$

on both of its GZ vectors. The X_k separate because they generate; Lecture 11 uses this.

10.5 The degenerate affine Hecke relations

The Coxeter generators $s_i = (i\ i{+}1)$ generate S_n (Proposition 2.20); no reflection s appears in this block. We compare their actions with the Jucys–Murphy operators.

Theorem 10.17 (Relations). *For $1 \leq i \leq n-1$ and $1 \leq k \leq n$,*

$$s_i X_i + 1 = X_{i+1} s_i, \qquad s_i X_k = X_k s_i \qquad (k \neq i, i+1).$$

Here 1 is the unit e of the group algebra.

Proof. *The adjacent relation.* For $j < i$, s_i fixes j, so by Theorem 2.16(i)

$$s_i (j\ i) s_i^{-1} = (j\ i{+}1).$$

Summing over $j < i$, as $s_i^{-1} = s_i$,

$$s_i X_i s_i = \sum_{j<i} (j\ i{+}1) = X_{i+1} - s_i.$$

Multiply on the right by s_i: $s_i X_i = X_{i+1} s_i - 1$.

The commutation relation. If $k < i$, s_i fixes $1, \dots, k$, so it commutes with each $(j\,k)$, $j < k$. If $k > i+1$, s_i fixes k and permutes $1, \dots, k-1$, so

$$s_i X_k s_i^{-1} = \sum_{j<k} (s_i(j)\ k) = X_k.$$ ∎

Multiplying the adjacent relation by s_i on both sides gives the twin $X_i s_i + 1 = s_i X_{i+1}$. The s_i also satisfy the *Coxeter relations*:

$$s_i^2 = e, \qquad s_i s_{i+1} s_i = (i\ i{+}2) = s_{i+1} s_i s_{i+1},$$
$$s_i s_j = s_j s_i \qquad (|i-j| \geq 2).$$

Here $1 \leq i \leq n-1$ in $s_i^2 = e$, $1 \leq i \leq n-2$ in the braid relation, and $1 \leq i, j \leq n-1$ in the distant relation. The braid equality follows by conjugating transpositions (Theorem 2.16(i)).

Definition 10.18 (The degenerate affine Hecke relations). Elements $s_1, \dots, s_{n-1}$ and $X_1, \dots, X_n$ of an algebra satisfy the *degenerate affine Hecke relations* if the s_i satisfy the Coxeter relations, the X_k commute pairwise, and the relations of Theorem 10.17 hold. Here e means the identity 1 of the algebra; the abstract braid relation is the equality of the two outer terms above, with $(i\ i{+}2)$ the concrete permutation in S_n.

So the s_i and X_k of $\mathbb{C}[S_n]$ satisfy them, with $X_1 = 0$. Lecture 11 starts from:

Corollary 10.19 (Two vectors). *Let v be a GZ vector of weight $(a_1, \dots, a_n)$, and $1 \leq i \leq n-1$. Then*

$$X_i\, s_i v = a_{i+1}\, s_i v - v, \qquad X_{i+1}\, s_i v = a_i\, s_i v + v,$$
$$X_k\, s_i v = a_k\, s_i v \qquad (1 \leq k \leq n,\ k \neq i, i+1).$$

So the span of v and $s_i v$ is invariant under s_i and every X_k.

Proof. Apply the twin, the adjacent relation and the commutation relation, in turn, to v. For instance,

$$X_i\, s_i v = (s_i X_{i+1} - 1)v = a_{i+1}\, s_i v - v.$$

And $s_i(s_i v) = v$, as $s_i^2 = e$. ∎

10.6 Worked example

We choose a GZ basis of E and compute the actions of X_2 and X_3.

The basis. In $\mathbb{C}^3$, the e_i are basis vectors, distinct from the group identity e. Take Lecture 5's $f_1 = e_1 - e_2$ and $f_2 = e_2 - e_3$ (§ 5.5), and $u = f_1 + 2f_2$, the vector $(1, 1, -2)$. Then

$$(1\,2) \cdot u = u, \qquad (1\,2) \cdot f_1 = -f_1.$$

So $\mathbb{C}u$ and $\mathbb{C}f_1$ are the $\mathbf{1}$- and sgn-isotypic components of $\mathrm{Res}^{S_3}_{S_2} E$, the lines of $\mathbf{1} \to \mathbf{1} \to E$ and $\mathbf{1} \to \mathrm{sgn} \to E$.

$X_3 = (1\,3) + (2\,3)$. On the two vectors,

$$X_3 u = (1\,3) \cdot u + (2\,3) \cdot u = (-2, 1, 1) + (1, -2, 1) = -u,$$
$$X_3 f_1 = (1\,3) \cdot f_1 + (2\,3) \cdot f_1 = (0, -1, 1) + (1, 0, -1) = f_1.$$

In the basis (u, f_1), with $X_2 = (1\,2)$,

$$D(X_2) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad D(X_3) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The weights are $(0, 1, -1)$ for u and $(0, -1, 1)$ for f_1. As Proposition 10.10 predicts, $a_3 = \omega_E(\tau_3) - \omega_{U_2}(\tau_2)$,

$$0 - 1 = -1 \quad (U_2 = \mathbf{1}), \qquad 0 - (-1) = 1 \quad (U_2 = \mathrm{sgn}).$$

Self-check. Check $s_2X_2 + 1 = X_3s_2$ on u. *Answer:* from the display above,

$$(2\,3)\cdot u = (1,-2,1) = -\frac{1}{2}u + \frac{3}{2}f_1.$$

As $X_2u = u$, the two sides give

$$(s_2X_2+1)u = s_2u + u = \frac{1}{2}u + \frac{3}{2}f_1,$$
$$X_3s_2u = -\frac{1}{2}X_3u + \frac{3}{2}X_3f_1 = \frac{1}{2}u + \frac{3}{2}f_1.$$

Summary.

- *The main results:* multiplicity-free branching for $n \geq 2$ (Theorem 10.3); Gelfand–Tsetlin bases and the maximal commutative GZ(n) (Theorem 10.6), generated by the Jucys–Murphy elements (Theorem 10.14), whose weights separate GZ vectors (Corollary 10.15); the relations (Theorem 10.17).
- *Handed on:* weights, Corollary 10.15, the relations and Corollary 10.19 to Lecture 11; the branching graph to Lecture 12.
- *Next:* Lecture 11 asks which weights occur.

References

Vershik and Okounkov, *A new approach to the representation theory of the symmetric groups. II* (arXiv:math/0503040): Prop. 1.1, p. 8; Thm 2.1, p. 9; Thms 2.5 and 2.8 and Cor. 2.6, pp. 10–12; (3.2)–(3.3), p. 14. Ceccherini-Silberstein, Scarabotti and Tolli, *Representation Theory of the Symmetric Groups*: §2.2, pp. 69–72; Thm 3.2.1, p. 94; Olshanskii's theorem, Thm 3.2.6, p. 97, and Cor. 3.2.7, p. 98.[1] Lorenz, *A Tour of Representation Theory*: §§4.1–4.2, pp. 189–196.

[1] They write T for a path.